

Plan B: when the two laptops cannot talk

Read this before the day, not during it. There are four fallbacks, each weaker than the last, and every one of them still gives you a working demo. The worst case is not “the demo fails”; it is “the demo runs and we describe honestly what is missing”.

First, know what actually broke

Three different things look identical from the outside. Spend twenty seconds telling them apart before reaching for a fallback.

TEST ON LAPTOP A	RESULT	WHAT IT MEANS
ping 192.168.137.132	replies	network is fine, the problem is the program or the firewall
ping 192.168.137.132	no reply	they are not on the same network, or the network blocks it
curl http://<ip>:8000/witness/pubkey	a key id	everything works. The fault is elsewhere
curl	connection refused	the witness program is not running on laptop B
curl	hangs, then times out	a firewall is blocking the port

THE MOST COMMON CAUSE BY FAR

The window running `python node.py` on laptop B was closed. Check that before anything else. If it is closed, start it again and you are done.

The four fallbacks, strongest first

PLAN B1

A different hotspot

Any phone will do, including a judge’s if you are brave. Turn on the hotspot, join both laptops, get the new address on laptop B with `ipconfig`, and re-run with that address.

```
python scripts/demo_bundle.py --witness http://<new-ip>:8000
```

Cost: about ninety seconds. Nothing is lost. Every claim still holds, including the two-machine signature.

PLAN B2**A cable between the two laptops**

If you have a USB-to-Ethernet adapter and a cable, connect the laptops directly. Windows gives each an address automatically. This needs no wireless at all and cannot be blocked by a venue network.

Check the address on both with `ipconfig` and use whatever laptop B reports. **Everything still holds.**

Plain USB-C to USB-C will *not* work between two laptops unless both have Thunderbolt. We tested this and recorded the result. Do not waste demo minutes on it.

PLAN B3**Show a record that was already co-signed**

You already have a real ledger, produced on 3 September 2026, carrying a genuine second signature from laptop B's key. It is saved at `D:\akhand SIH\demo-data\chain.json`. Open it on the screen and run the live check.

What still works: the whole integrity demonstration. The browser recomputes every fingerprint, the tamper demo names the exact broken record, the certificate renders with its clauses, and the co-signer key id is visible on the first row.

What you must say:

"The two machines are not talking right now, so this is a record we produced earlier today rather than one made in front of you. The second signature in it is real, from a key that is not on this laptop. What you are watching the browser do is live: it is recomputing every fingerprint now."

That is an honest and strong position. The verification is genuinely live; only the *creation* of a fresh record is not.

PLAN B4**Run without the witness, and let it refuse**

If laptop B is dead entirely, run the tool alone:

```
python scripts/demo_bundle.py
```

Records are then marked SOFTWARE_KEY rather than WITNESS_COSIGNED. The tool has honestly downgraded the label because the second party was absent.

THIS FAILURE IS ITSELF A DEMONSTRATION

Point at the tier on screen and say it out loud:

“The second computer is not available, so the tool has recorded the weaker trust level. It has not pretended. That is the behaviour we are asking you to judge: it would rather tell you it had less help than claim it had more.”

Then run the deliberate refusal so they see a blocked operation written into the record. A judge who sees a tool refuse itself in public remembers it.

Fixing the firewall, if that is the cause

Symptom: ping works but curl times out. Laptop B is blocking the port. On **laptop B**, open PowerShell as Administrator (right-click, Run as administrator):

```
New-NetFirewallRule -DisplayName "Akhandas witness 8000" `
  -Direction Inbound -LocalPort 8000 -Protocol TCP -Action Allow
```

Or set the network to Private, which Windows treats far more permissively:

```
# see which profile the hotspot connection is on
Get-NetConnectionProfile
# then set it, using the name shown above
Set-NetConnectionProfile -Name "<name>" -NetworkCategory Private
```

DO THIS THE NIGHT BEFORE

The rule persists. Adding it once at home means it can never cost you demo minutes.

How to make Plan B unnecessary

DO THIS BEFORE THE DAY	WHY
Copy the witness folder onto laptop B's own hard disk	if it runs from a USB stick, unplugging the stick kills it
Add the firewall rule above, once	removes the most common silent failure
Fix laptop B's address, or write it down each morning	addresses change overnight when the lease renews
Charge both laptops and bring one power strip	the venue gives one socket per team
Bring a second phone for a spare hotspot	a flat battery should not end the demo
Keep the pre-made record in demo-data\	this is Plan B3, and it costs nothing to carry
Rehearse the tamper and restore once, out loud	so the words are ready when the room is watching

THE THING THAT MAKES ALL OF THIS SURVIVABLE

The verification runs **inside the browser**, on the machine in front of you. It never needed a network, a server, or the second laptop to prove that a record is intact. The second laptop matters when a record is *created*, not when it is *checked*. So even in the worst case, the most convincing part of the demonstration still works exactly as designed.

The thirty-second decision

SITUATION	DO THIS	WHAT YOU LOSE
Witness window closed	restart node.py	nothing
Hotspot dropped	Plan B1, another phone	nothing
Venue blocks laptop-to-laptop	Plan B1 or B2	nothing
Firewall blocking, no admin rights	Plan B3, the saved record	only live creation
Laptop B will not boot	Plan B3, then B4	only live creation
Under two minutes left	Plan B3 immediately	nothing that matters

NEVER DO THIS

Do not describe a saved record as one made live. Do not claim a co-signature the run did not have. Do not edit a record to make the screen look better. The entire argument of this project is that the tool does not overstate what happened, and a team that overstates in the room has lost the argument before the questions begin.

Akhanda · Syntax Squad · GITAM (Deemed to be University), Hyderabad · Smart India Hackathon 2026, SIH26149, sponsored by the National Technical Research Organisation. Keep this page with the laptops, not on them.